

Large language models as labeling oracles for 621-category product classification in Portuguese: an instrumented evaluation of quality and cost

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Abstract

Large language models (LLMs) are increasingly used as labeling oracles, but the evidence base concentrates on tasks with tens to a few hundred classes and rarely treats cost or the measurement instrument as objects of study. We evaluate six LLM oracles on zero-shot classification of short Portuguese retail product descriptions into a closed space of 621 categories — well beyond prior product-classification studies — under a fully instrumented protocol: output constrained to the label space, batch annotation with paired calibration, cache-aware cost accounting, serving provenance, and paired statistics (Wilson intervals, exact McNemar). Four oracles plateau at 77–83% accuracy, ≈ 7 points under a supervised ceiling trained on 250,000 labels, at US\$ 0.00–0.92 per thousand labels; the best-value model delivers 95% of the top accuracy at 8.5% of its cost, and a zero-cost oracle is statistically tied with it. Plateau Macro-F1 *exceeds* a fully supervised light baseline, because LLMs are strongest precisely in rare classes. Nearly half of oracle error reflects catalog conventions rather than product misunderstanding. Two findings concern the instrument itself: without output constraints, 6.8% of responses become spurious phrasing errors charged against the model; and the same model served by two providers disagrees with itself ($p < 0.001$) on identical instances. A prompt ablation with boundary rules improves the production distribution (+4.6 points, $p = 0.012$) while severely degrading the stratified tail (−10.8 points, $p < 0.001$). Code, prompts and per-instance annotation artifacts are released.

Keywords: large language models; data annotation; text classification; short text; active learning; annotation cost; Portuguese.

1 Introduction

Supervised text classification remains bounded by the cost of labeled data. The recent proposal to replace human annotators with large language models promises orders-of-magnitude cost reduction (Gilardi et al., 2023), and a rapidly growing literature explores LLMs as annotators inside active learning loops (Zhang and Takada, 2025; Rouzegar and Makrehchi, 2024; Kholodna et al., 2024; Bayer and Reuter, 2024). Yet three aspects of this evidence base limit its transfer to production settings. First, *granularity*: published product-classification evaluations operate on tens to ≈ 370 categories (Roumeliotis et al., 2025; Gholamian et al., 2024); industrial catalogs

routinely exceed several hundred. Second, *cost*: most studies report label counts, not monetary cost, and none we are aware of instruments the pricing levers that dominate real spending — batch amortization of the prompt prefix and provider-side prefix caching. Third, and least discussed, the *measurement instrument*: accuracy numbers are reported as properties of a model, when they are in fact properties of a (model, prompt, output contract, serving provider, date) configuration.

This paper evaluates six LLM oracles — spanning commercial APIs, MaaS platforms and a zero-cost evaluation tier — on zero-shot classification of Brazilian retail product descriptions (4–50 characters, uppercase, aggressive abbreviations) into a closed space of **621 categories**. The evaluation is *instrumented* end to end: every annotation records its instrument mode, batch size, token counts and serving provenance; all comparisons are paired on identical instances; and the gold standard’s own error rate is audited and propagated as a measurement ceiling.

We ask four research questions. **RQ1**: what accuracy and Macro-F1 do LLM oracles reach at this granularity, and are their differences statistically real? **RQ2**: what does a thousand labels actually cost, and how do batching and caching change it? **RQ3**: what is the anatomy of oracle error? **RQ4**: how much of a measured “error rate” is an artifact of the measurement instrument itself?

Contributions: (i) the highest-granularity instrumented evaluation of LLM-as-annotator for product classification to date, in a lower-resource language; (ii) a reusable measurement protocol (output contract, paired batch calibration, cache-aware cost accounting, provenance triple) with public artifacts; (iii) two instrument-level findings — the output-format artifact and serving-provider divergence — that affect any future oracle evaluation; and (iv) an actionable error anatomy plus a prompt-ablation result showing that boundary rules derived from aggregate error profiles transfer asymmetrically between the production and tail distributions.

2 Related work

LLMs as annotators. Gilardi et al. (2023) established that zero-shot ChatGPT can outperform crowdworkers at a fraction of the cost; subsequent work integrated LLM annotation into active learning loops (Zhang and Takada, 2025), routed hard cases to humans by confidence (Rouzegar and Makrehchi, 2024), and demonstrated batch annotation economics in low-resource languages (Kholodna et al., 2024). This line inherits the older noisy-annotator tradition (Snow et al., 2008; Sheng et al., 2008); a known caveat is that LLM self-reported confidence tends toward overconfidence and requires calibration (Tian et al., 2023), which is why our protocol relies on external measurement rather than model-reported certainty. Evaluations at product-catalog scale are rarer: Roumeliotis et al. (2025) compare zero-shot LLMs on 248 categories; Gholamian et al. (2024) test robustness on up to 370 leaf categories. Our 621-way closed space extends this frontier while adding paired statistics and cost instrumentation absent from both.

Annotation noise. Oracle errors become label noise for any downstream learner; taxonomies and robustness results (Frénay and Verleysen, 2014; Song et al., 2023) distinguish uniform from structured noise, and recent benchmarks show *real* annotation noise to be more harmful than synthetic corruption (Merdjanovska et al., 2024; Rączkowska et al., 2024). Our error anatomy (RQ3) characterizes where LLM oracle noise falls in this spectrum.

Portuguese retail classification. Machado and Veloso (2026) classify $\approx 100k$ Portuguese supermarket titles into ECOICOP categories with fine-tuned BERT variants, relying on an

ad-hoc human-labeling pipeline; they do not evaluate LLM oracles. Community surveys report deployment complexity and cost uncertainty as the main barriers to LLM-assisted annotation (Romberg et al., 2025) — precisely the gap an instrumented cost protocol addresses.

3 Task, data and gold standard

The task is single-label classification of short product descriptions from Brazilian fiscal receipts (e.g., CERV BRAHMA LT 350ML) into a closed catalog of 621 product-type categories. The dataset comprises ≈ 250 k labeled descriptions (Darú, 2024); texts are 4–50 characters, uppercase, with domain-specific abbreviations. The label space exhibits severe long-tail imbalance, which motivates Macro-F1 as the primary quality metric alongside accuracy.

Gold-standard audit. Since the reference labels were produced by a human process, we audited label conflicts (identical normalized text with divergent labels) and re-scored all oracles (a) excluding conflicting instances and (b) under multi-gold scoring. Accuracy shifts by at most +0.7 points and no ranking changes; the known gold error bounds the measurement ceiling at $\approx 99.3\%$ and is reported alongside all results rather than ignored.

Evaluation samples. All oracles annotate *the same* instances (fully paired design). **S-rand** ($n = 1,000$, simple random) preserves the production distribution and serves RQ1/RQ2/RQ4; its size caps the Wilson 95% half-width at ≈ 3 points (2.5 in the observed plateau). **S-strat** (3 per class, $n = 1,863$) provides per-class support for Macro-F1 and confusion analysis (RQ3): with 621 classes, a random sample leaves most classes unrepresented, and three instances per class is the minimum that separates systematic (0/3, 3/3) from sporadic (1/3, 2/3) class error.

4 Measurement instrumentation

Output contract. Three instrument modes are defined and recorded per annotation: **enum** (production mode) — structured output with the label constrained at decoding time to the 621-value schema (provider-enforced structured outputs); **json-prompt** — for providers without structured output support, the category list and JSON format are instructed in the prompt, with strict post-validation and explicit accounting of invalid labels; **free** — structured output *without* label constraint, a controlled replica of the naive instrument used solely to quantify its artifact (RQ4).

Prompt and batching. The prompt contextualizes the domain (fiscal-receipt abbreviations), instructs type-level classification ignoring brand and packaging, and discourages misuse of the sentinel rare-class label. The static prefix (instructions + schema) precedes the variable items, exploiting provider prefix caches. Annotation is batched (numbered items in one call, indexed vector response); batch size is calibrated in a paired pre-study (1 vs. 10 vs. 25 items on identical instances): exact McNemar detects no degradation ($b=1$ vs. $b=10$: $p = 0.58$), while batching cuts cost per label by up to 10 $\times$ through prefix amortization — consistent with the batch-annotation economics of Kholodna et al. (2024). We adopt $b = 10$ (OpenAI) and $b = 25$ (MaaS), recorded per annotation.

Instrument-mode assignment and scoring convention. The output contract is dictated by provider capability: enum for providers with structured-output support (OpenAI), json-prompt elsewhere. This is a potential confound between provider families, which the protocol handles by (a) reporting invalid-label rates separately in every table, (b) scoring invalid outputs as errors — so json-prompt oracles are penalized, not shielded, by the weaker contract — and (c) quantifying the maximal size of the contract effect directly in RQ4 with the same model under both contracts. Cost is computed from provider-metered token counts per call multiplied by the provider price sheet at execution date, with cached prefix tokens billed at the provider’s cached rate; all per-call token counts are in the released artifacts.

Provenance and statistics. Every annotation records the (model, provider, mode, batch, temperature, date) tuple; temperature is 0 throughout, which reduces but does not eliminate serving nondeterminism — each cell is a single annotation run, a limitation stated in Section 6. Accuracies carry Wilson 95% intervals (Wilson, 1927; Brown et al., 2001); model pairs are compared with McNemar’s test on discordant pairs (exact binomial when $b + c < 25$) (McNemar, 1947; Dietterich, 1998), the recommended regime for classifiers evaluated once on a common sample. Two comparisons were pre-registered as decision-relevant (best vs. runner-up for the advanced-oracle role; zero-cost vs. cheapest-paid for the initial-oracle role); remaining pairwise tests are reported as exploratory, and no result in this paper depends on a marginal p -value surviving multiplicity correction.

5 Results

5.1 RQ1: quality at 621-way granularity

Table 1 reports the paired evaluation. Four oracles form a plateau at 77–83% accuracy on both samples. Within-plateau differences are small but partly real: deepseek-v4-pro is significantly better than v4-flash on S-strat (McNemar $b=43$, $c=16$, $p < 0.001$) and statistically tied with gpt-4o ($p = 0.061$). The small gpt-4o-mini collapses on the stratified sample (45.7%): its errors concentrate systematically in rare classes, which S-strat exposes by construction. No oracle reaches the 85% pre-registered adoption gate; the best zero-shot result sits ≈ 7 points below a fully supervised ceiling trained on 250k labels (89.6%, Darú, 2024). Conversely, plateau Macro-F1 (0.78–0.80 on S-strat) *exceeds* the 0.70 of a light supervised model trained on the full label budget: world knowledge compensates exactly where supervision is scarcest — the long tail.

5.2 RQ2: cost, cache and the serving provider

Costs within the same accuracy plateau span more than an order of magnitude (Table 2): deepseek-v4-flash delivers 78–82% at US\$ 0.035/1k — $26\times$ cheaper than gpt-4o at comparable quality — and two to three orders of magnitude below typical human annotation costs (Gilardi et al., 2023). Prefix caching reaches 88–95% hit rates where supported, halving effective input-token cost. Cost, however, is not the only operational constraint: provider throughput limits (3 requests/min on the MaaS tier; 18/min on a free aggregator tier) shape wall-clock time as much as price — a cheap, slow oracle can be dominated by an expensive, parallel one when the deadline binds.

The zero-monetary-cost regime proved competitive: nemotron-3-ultra served via NVIDIA’s evaluation API ties deepseek-v4-flash on S-rand (McNemar, $p = 0.76$). Two operational caveats are findings in themselves. (i) Free is not operationally free: a first attempt through an aggregator

Table 1: Oracle performance on paired samples (temperature 0, calibrated batching). Enum mode for OpenAI; json-prompt for MaaS/NIM. CI: Wilson 95%.

Oracle	Sample	Acc.	95% CI	Macro-F1	Invalid
deepseek-v4-pro	S-rand	82.1%	[79.6, 84.4]	0.785	0.1%
gpt-4o	S-rand	80.9%	[78.4, 83.2]	0.788	0.0%
deepseek-v4-flash	S-rand	78.3%	[75.6, 80.8]	0.750	0.7%
nemotron-3-ultra (free)	S-rand	77.9%	[75.2, 80.4]	0.752	1.2%
glm-5.2	S-rand	77.3%	[74.6, 79.8]	0.742	0.0%
gpt-4o-mini	S-rand	61.3%	[58.2, 64.3]	0.518	0.0%
deepseek-v4-pro	S-strat	82.6%	[80.8, 84.2]	0.793	0.0%
gpt-4o	S-strat	82.1%	[80.3, 83.8]	0.797	0.0%
deepseek-v4-flash	S-strat	81.6%	[79.7, 83.3]	0.781	0.9%
glm-5.2	S-strat	80.9%	[79.1, 82.6]	0.782	0.0%
nemotron-3-ultra (free)	S-strat	79.6%	[77.7, 81.4]	0.770	2.6%
gpt-4o-mini	S-strat	45.7%	[43.4, 48.0]	0.427	0.0%

Table 2: Cost per thousand labels (US\$, prices at execution, Jul/2026) and prefix-cache hit rate on S-rand.

Oracle	US\$/1k	Cache	Acc.
nemotron-3-ultra (free)	0.000	—	77.9%
deepseek-v4-flash	0.035	—	78.3%
gpt-4o-mini	0.051	91%	61.3%
glm-5.2	0.249	—	77.3%
deepseek-v4-pro	0.410	—	82.1%
gpt-4o	0.923	88%	80.9%

with a 50-requests/day quota projected six days of wall time for the same workload. (ii) *The serving provider is part of the instrument*: the same nemotron model served by the aggregator disagrees significantly with itself served by NVIDIA on identical instances (paired subset $n = 850$, $p < 0.001$). Oracle measurements should therefore record not only the model, but who serves it and when — each number is a photograph of a (model, provider, date) triple, not a property of model weights.

5.3 RQ3: anatomy of oracle error

Errors of the best oracle (v4-pro, S-strat; $n = 324$) were categorized by scripted lexical rules (shared head-word between predicted and gold label; umbrella-class markers; sentinel/invalid detection) with manual spot-checking; the categorization script and the labeled error list are in the released artifacts. The result: **31%** are confusions between *sibling categories* sharing a head word (*antiséptico bucal* $\rightarrow$ *enxaguante bucal*); **17%** have an umbrella class (“other . . .”) as gold while the model picks the specific class, and 7% the reverse; $\approx 2\%$ involve the sentinel class or invalid output; the remainder are genuine semantic errors. Nearly half of oracle error therefore reflects not product misunderstanding but ignorance of *this catalog’s conventions* — boundaries between neighbors and aggregation policy, learnable only from labeled examples. Two consequences: it motivates the prompt ablation of Section 5.5, and it predicts oracle noise is *structured* (concentrated in adjacent pairs), the more benign regime for downstream learners

(Frénay and Verleysen, 2014; Song et al., 2023).

5.4 RQ4: the instrument effect

Holding model (gpt-4o-mini), instances (S-rand) and prompt constant, and switching only the output contract — enum vs. free — yields accuracy 61.3% vs. 60.5%, but invalid-label rates of **0.0% vs. 6.8%**. The instrument’s effect appears less in point accuracy than in the *accounting of error*: without output constraints, nearly 7% of responses are out-of-space phrasings that a naive protocol counts as model error, deflating measurements and contaminating comparisons between models with different paraphrasing propensities. Enum-constrained decoding with strict validation eliminates this artifact class by construction.

5.5 Prompt as an instrument variable

Since $\approx 48\%$ of RQ3 errors are convention errors, we test whether prompt engineering can recover them in the cheapest cached model (gpt-4o-mini): variant **v4a** adds ten explicit boundary rules derived from the aggregate error profile; **v4b** adds ten *invented* boundary examples (never sampled from evaluation data); both are paired against the base prompt on the first 500 items of each sample — a prefix subset (random by construction, as samples are pre-shuffled) sized to detect shifts of 4–5 points via McNemar given the expected 60–80 discordant pairs, at one fifth of the full-sample cost per variant. On S-rand, rules recover part of the convention error: +3.8 points (v4a, $p = 0.045$) and +4.6 points (v4b, $p = 0.012$). On S-strat the same rules *destroy* performance: -7.0 and -10.8 points ($p < 0.001$), with invented examples significantly worse than rules alone ($p = 0.0013$). Rules distilled from aggregate error profiles push the model toward the conventions of *frequent* classes — exactly the behavior the stratified sample punishes. Prompt optimization for labeling oracles must therefore be validated on both the production and the tail distribution before adoption; a gain measured only on the natural distribution can mask severe tail regression.

6 Discussion

Choosing an oracle is a portfolio decision. At 621-way granularity no oracle passes a 85% adoption gate, yet the plateau’s Macro-F1 beats a fully supervised light baseline. The practical reading for label-scarce pipelines: the LLM oracle is not a replacement for supervision but a complementary specialist for the long tail, at US\$ 0.00–0.41 per thousand labels. The $26\times$ price spread inside a statistical tie makes oracle selection a measurable engineering decision, not a brand choice — and our protocol makes it auditable.

Measure the instrument, not just the model. Two of our findings would silently bias any naive replication: the 6.8% output-format artifact and the significant self-disagreement across serving providers. Both argue that credible LLM-annotation papers should publish their output contract, serving provenance and per-instance artifacts — as we do — and that leaderboard-style comparisons without this metadata are fragile.

Threats to validity. Single dataset and language pair (mitigated by public artifacts and a replication design on AlleNoise (Rączkowska et al., 2024) as future work); a single annotation run per cell — temperature 0 reduces but does not eliminate serving nondeterminism, and the RQ2

provider-divergence finding quantifies exactly why this matters; API prices dated at execution (ratios are more stable than absolute values); boundary rules in the ablation derive from this catalog’s error profile (the method generalizes; the rules do not); gold-standard noise bounds all accuracies at $\approx 99.3\%$, audited and propagated.

7 Conclusion

We provide the highest-granularity instrumented evaluation of LLMs as labeling oracles to date: 621 closed categories, Portuguese short text, six oracles across three cost regimes, fully paired statistics, and cache-aware cost accounting. LLM oracles plateau at 77–83%, seven points under a 250k-label supervised ceiling, while beating it on Macro-F1; a zero-cost oracle ties the best cost-efficiency tier; nearly half of oracle error is catalog convention, recoverable by prompt rules only at the tail’s expense; and the measurement instrument itself — output contract and serving provider — moves results by margins that exceed several model-to-model differences. The protocol and artifacts are public and directly reusable for oracle selection in any closed-taxonomy annotation pipeline.

Artifacts. Code, prompts, per-instance annotations, token counts and analysis scripts: github.com/GHDaru/active-llm-experiments (e0, e0p). The underlying corpus is publicly available on Kaggle (DOI 10.34740/kaggle/dsv/426534). The paired evaluation samples and all annotation artifacts are released with this paper. [TODO: DOI/Zenodo release do código antes da submissão]

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